

Varsha Kumbenahally Roopeshkumar

vk2772@nyu.edu | (646) 554-5215 | [LinkedIn](#) | [Website](#) | [GitHub](#)

EDUCATION

New York University

New York, NY

Master of Science in Cybersecurity

Expected May 2027

- Coursework in Network Security, Applied Cryptography, Application Security, Cloud Computing, and Information Security and Privacy
- Active in the OSIRiS Cybersecurity Club, solving CTF challenges across AES oracle attacks, RSA exploitation, buffer overflows, SQL injection, and binary exploitation

PES University

Bengaluru, India

B.Tech in Computer Science

May 2024

- Coursework in Applied Cryptography, Ethical Hacking, Blockchain, Network Security, and Data Structures and Algorithms

TECHNICAL SKILLS

- **Languages:** Python, C, C++, Java, SQL, JavaScript, Node.js, MATLAB, R
- **Security Tools:** Burp Suite, Metasploit, Wireshark, Nmap, Kali Linux, Splunk, pwntools
- **Cryptography:** AES-256-GCM, RSA-OAEP, RSA-PSS, ECDSA, MACs, digital signatures, key management
- **Cloud and DevOps:** AWS (Bedrock, CloudWatch, Lambda), Docker, Kubernetes, Jenkins, Git
- **Standards:** ISO/SAE 21434, AUTOSAR SecOC
- **CTF Skills:** Buffer overflow, SQL injection, XOR and AES oracle attacks, Wiener attack on RSA, command injection

PROFESSIONAL EXPERIENCE

Course Assistant, K-12 STEM Education

New York, NY | Jun 2026 - Aug 2026

- Teach cybersecurity fundamentals to high school students in a K-12 STEM summer program
- Build hands-on lessons and labs that make core security concepts approachable for beginners

Cybersecurity Associate Software Engineer, APTIV

Bengaluru, India | Apr 2025 - Jun 2025

- Hardened radar ECUs across the SRR500 to SRR700 line with AES-256 encryption, RSA authentication, digital signatures, and MACs, aligned with ISO/SAE 21434
- Embedded cryptographic controls into the AUTOSAR stack for secure over-the-air updates and used Python to automate security validation checks

Project Intern, APTIV

Bengaluru, India | Jan 2024 - Jun 2024

- Programmed Jacinto J6 and RH850 microcontrollers over SPI and UART for an in-vehicle infotainment system
- Added secure coding, encryption, and software update handling to protect the embedded system against threats

SECURITY PROJECTS

LLM Sentinel, AI Prompt Injection Defense | AWS, Python, Bedrock, Streamlit

2026

- Designed a three-layer cloud defense (Edge Filter, Semantic Guard, Egress Auditor) that protects an AI chatbot against prompt injection
- Added real-time CloudWatch alerting and a Streamlit dashboard so threats stay visible as they happen

SecureVault, End-to-End Encrypted Cloud Storage | Python, AES-256-GCM, RSA-2048, Google Drive API

2026

- Built hybrid-encryption cloud storage with RSA-OAEP key encapsulation, RSA-PSS signatures, and replay-attack protection
- Shipped both CLI and GUI interfaces and tested the system against tampering, spoofing, and replay attacks

Kubernetes Deployment Pipeline | Kubernetes, Docker, Minikube, Prometheus, Grafana

2026

- Deployed a containerized Flask and MongoDB to-do app to Kubernetes on Minikube with two replicas
- Set up self-healing replicas, zero-downtime rolling updates, health probes, and Prometheus alerts to Slack

Automotive Cybersecurity, APTIV | AES, RSA, Digital Signatures, AUTOSAR, G-Test

2025

- Secured ADAS and ABS systems and radar-based object detection using AES encryption, asymmetric key signing, and digital signatures
- Built key management for secure data storage and ran G-Test unit tests to verify the security of connected-vehicle systems

PUBLICATIONS

- A Comprehensive Study of Load Balancing Approaches in Real-Time Multi-Core Systems. IEEE Access, vol. 12, 2024. [DOI: 10.1109/ACCESS.2024.3388291](#)
- Load Balancing Based Scheduling for Mixed Real-Time Tasks on Multicore Systems. IEEE PDGC, 2024. [DOI: 10.1109/PDGC64653.2024.10984400](#)